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Memory device and method of forming the same

Abstract

Some embodiments relate to a memory device. The memory device includes a transistor having a first source/drain (S/D) region and a second S/D region, a first S/D contact disposed over the first S/D region, the first S/D contact extending lengthwise in a first direction, a second S/D contact disposed over the second S/D region, a first via landing on the first S/D contact, the first via extending lengthwise in a second direction different from the first direction, a second via landing on the second S/D contact, the first via having a length measured in the second direction that is larger than the second via, a first conductive line coupled to the first via, a second conductive line coupled to the second via, and a memory structure disposed above the transistor and coupled to the second conductive line.

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Background/Summary

PRIORITY (1) This application claims the benefits to U.S. Provisional Application No. 63/282,880, filed Nov. 24, 2021, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of processing and manufacturing ICs.

(2) One advancement in some IC design and fabrication has been the developing of non-volatile memory (NVM), and in particular to magnetic random-access memory (MRAM). MRAM offers comparable performance to volatile static random-access memory (SRAM) and comparable density with lower power consumption to volatile dynamic random access memory (DRAM). Compared to NVM Flash memory, MRAM may offer faster access times and suffer less degradation over time. An MRAM cell is formed by a magnetic tunneling junction (MTJ) comprising two ferromagnetic layers which are separated by a thin insulating barrier, and operates by tunneling of electrons between the two ferromagnetic layers through the insulating barrier. In operation, the variable states (e.g., logical “0” or “1” state) of an MRAM cell is typically read by measuring the resistance of the MTJ. Due to magnetic tunnel effect, the resistance of the MTJ changes with the variable magnetic polarity. When a voltage bias is applied across a combined structure of a top metal line (e.g., a bit line), a MTJ, a control transistor configured to drive the MTJ, and a bottom metal line (e.g., a common source line), one can obtain a series resistance of the combined structure when a current flowing therethrough is measured. The series resistance includes the resistance of the MTJ and additional resistance. The additional resistance shall be reduced to or kept at a desirable value as low as possible in order to improve sensitivity and speed of the MRAM cell. Although existing approaches in MRAM device formation have generally been adequate for their intended purposes, they have not been entirely satisfactory in all respects. For example, routing resistance associated with the control transistor is an unneglectable contributor to the additional resistance in an MRAM cell and may degrade memory circuit performance if its value is high. Accordingly, there exists a need for improvements in this area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale and are used for illustration purposes only. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

- (2) FIG. 1 is a diagram of a memory system, in accordance with some embodiments.
- (3) FIG. 2 is a perspective view of a memory cell, in accordance with some embodiments.
- (4) FIG. 3 illustrates a schematic view of a memory array, in accordance with some embodiments.
- (5) FIG. 4 illustrates a cross-sectional view of a semiconductor device with a memory array comprising MTJs, in accordance with some embodiments.
- (6) FIG. 5 illustrates a top view of a semiconductor device with a memory array comprising MTJs, in accordance with some embodiments.
- (7) FIGS. 6A, 6B, 6C, 7A, and 7B illustrate cross-sectional views of the semiconductor device in FIG. 5, in accordance with some embodiments.
- (8) FIG. 8 illustrates a flow chart for a method of forming a semiconductor device with a memory array comprising MTJs, in accordance with some embodiments.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. Still further, when a number or a range of numbers is described with “about,” “approximate,” and the like, the term encompasses numbers that are within certain variations (such as $\pm 10\%$ or other variations) of the number described, in accordance with the knowledge of the skilled in the art in view of the specific technology disclosed herein, unless otherwise specified. For example, the term “about 5 nm” may encompass the dimension range from 4.5 nm to 5.5 nm, 4.0 nm to 5.0 nm, etc.

(11) The present disclosure is generally related to semiconductor devices and fabrication methods. More particularly, the present disclosure is related to providing a semiconductor device with an array of magnetic random-access memory (MRAM) devices (or cells) where the routing resistance associated with the transistors configured to control respective magnetic tunneling junctions (MTJs) is reduced. In some embodiments of the present disclosure, MTJs are disposed within metallization layers of a multi-layer interconnect (or MLI). The MTJs are coupled to respective control transistors for read/write control. Vias connecting source contacts of the transistors to conductive lines of a bottom metallization layer of the MLI (e.g., M0) are formed as a rail, which expands contact area between the vias and the source contacts, as well as contact area between the vias and the conductive lines, and in turn reduces routing resistance. By reducing the routing resistance, sensitivity and speed of the MRAM device is improved.

(12) FIG. 1 is a diagram of a memory system **100**, in accordance with some embodiments. The memory system **100** includes a memory controller **105** and a memory array **120**. The memory array **120** may include a plurality of storage circuits or memory cells **125** arranged in two- or three-dimensional arrays. Each memory cell **125** may be coupled to a corresponding word line WL and a

corresponding bit line BL. The memory controller **105** may write data to or read data from the memory array **120** according to electrical signals through word lines WL and bit lines BL. In other embodiments, the memory system **100** includes more, fewer, or different components than shown in FIG. 1.

(13) The memory array **120** is a hardware component that stores data. In one aspect, the memory array **120** is embodied as a semiconductor memory device. The memory array **120** includes a plurality of storage circuits or memory cells **125**. The memory array **120** includes bit lines BL0, BL1 . . . BLK, each extending in a first direction (e.g., X-direction) and word lines WL0, WL1 . . . WLJ, each extending in a second direction (e.g., Y-direction). The word lines WL and the bit lines BL may be conductive metals or conductive rails. In one aspect, each memory cell **125** is coupled to a corresponding word line WL and a corresponding bit line BL, and can be operated according to voltages or currents through the corresponding word line WL and the corresponding bit line BL. In one aspect, each memory cell **125** includes cross-coupled transistors and MTJs. Each memory cell **125** may be magnetic random-access memory (MRAM) memory cell with an MTJ. In some embodiments, the memory array **120** includes additional lines (e.g., select lines, reference lines, reference control lines, power rails, etc.).

(14) The memory controller **105** is a hardware component that controls operations of the memory array **120**. In some embodiments, the memory controller **105** includes a bit line controller **112**, a word line controller **114**, and a timing controller **110**. In one configuration, the word line controller **114** is a circuit that provides a voltage or a current through one or more word lines WL of the memory array **120**, and the bit line controller **112** is a circuit that provides or senses a voltage or current through one or more bit lines BL of the memory array **120**. In one configuration, the timing controller **110** is a circuit that provides control signals or clock signals to synchronize operations of the bit line controller **112** and the word line controller **114**. The bit line controller **112** may be coupled to bit lines BL of the memory array **120**, and the word line controller **114** may be coupled to word lines WL of the memory array **120**. In one example, to write data to a memory cell **125**, the word line controller **114** provides a voltage or current to the memory cell **125** through a word line WL coupled to the memory cell **125**, and the bit line controller **112** applies a bias voltage to the memory cell **125** through a bit line BL coupled to the memory cell **125**. In one example, to read data from a memory cell **125**, the word line controller **114** provides a voltage or current to the memory cell **125** through a word line WL coupled to the memory cell **125**, and the bit line controller **112** senses a voltage or current corresponding to data stored by the memory cell **125** through a bit line BL coupled to the memory cell **125**. In some embodiments, the memory controller **105** includes more, fewer, or different components than shown in FIG. 1.

(15) FIG. 2 illustrates a perspective view of an example memory cell **125** as a building block of the memory array **120** as shown in FIG. 1. Particularly, FIG. 2 illustrates a memory cell **125** that is an MRAM cell having an MTJ **180** (or MTJ stack **180**). The MTJ **180** includes an upper magnetic plate **182** (or top magnetic plate) and a lower magnetic plate **184** (or bottom magnetic plate), which are separated by a thin insulating layer **186**, also referred to as a tunnel barrier layer. One of the two magnetic plates (e.g., the lower magnetic plate **184**) includes a magnetic layer that is pinned (thus referred to as a pinned layer or a reference layer) to an antiferromagnetic layer (referred to as a pinning layer), while the other magnetic plate (e.g., the upper magnetic plate **182**) is a “free” magnetic layer (also referred to as a free layer) that can have its magnetic field changed to one of two or more values to store one of two or more corresponding data states.

(16) The MTJ **180** uses tunnel magnetoresistance to store magnetic fields on the upper and lower magnetic plates **182** and **184**. For a sufficiently thin insulating layer **186** (e.g., about 10 nm or less thick), electrons can tunnel from the upper magnetic plate **182** to the lower magnetic plate **184**. Data may be written to the cell in many ways. In one method, current is passed between the upper and lower magnetic plates **182** and **184**, which induces a magnetic field stored in the free layer (e.g., the upper magnetic plate **182**). In another method, spin-transfer-torque (STT) is utilized,

wherein a spin-aligned or polarized electron flow is used to change the magnetic field within the free layer with respect to the reference layer. Other methods to write data may be used. However, all data write methods include changing the magnetic field within the free layer with respect to the reference layer.

(17) The electrical resistance of the MTJ **180** changes in accordance with the magnetic fields stored in the upper and lower magnetic plates **182** and **184**, due to the magnetic tunnel effect. For example, when the magnetic fields of the upper and lower magnetic plates **182** and **184** are in the same direction (parallel), the MTJ **180** is in a low-resistance state (i.e., a logical “0” state). The resistance of the MTJ **180** under the low-resistance state is denoted as R_p . When the magnetic fields of the upper and lower magnetic plates **182** and **184** are in opposite directions (anti-parallel), the MTJ **180** is in a high-resistance state (i.e., a logical “1” state). The resistance of the MTJ **180** under the high-resistance state is denoted as R_{ap} . The direction of the magnetic field of the upper magnetic plate **182** can be changed by passing a current through the MTJ **180**. By measuring the resistance R_p or R_{ap} between the upper and lower magnetic plates **182** and **184**, a read circuitry coupled to the MTJ **180** can discern between the “0” and “1” states.

(18) FIG. 2 further shows that the upper magnetic plate **182** of an MTJ **180** is coupled to a bit line (BL), the lower magnetic plate **184** of an MTJ **180** is coupled to a drain (or source) of a transistor **190**, the source (or drain) of the transistor **190** is coupled to a source line (SL), and the gate of the transistor **190** is coupled to a word line (WL). That is, the MTJ **180** is sandwiched between the metal grids of word lines and source lines. The MTJ **180** can be accessed (such as read or written) through the bit line and the source line. When data is written to or read from the memory cell **125**, a word line is asserted to turn on the transistor **190**, and an appropriate bias is applied to a bit line to write or read respective value to or from the respective memory cell **125**. Driven by the appropriate bias, a current flowing through a combined structure of the bit line, the MTJ **180**, the transistor **190**, and the source line is measured. One can thus obtain a series resistance of the combined structure from values of the bias and current and derive the resistance of the MTJ **180**. When the routing resistance, denoted as R_s , from the bit line to the drain (or source) of the transistor **190** and from the source line to the source (or drain) of the transistor **190** is not ideal, the derived resistance of the MTJ **180** is actually the low resistance of the MTJ itself plus the routing resistance (i.e., $R_p + R_s$) under the low-resistance state and the high resistance of the MTJ itself plus the routing resistance (i.e., $R_{ap} + R_s$) under the high-resistance state.

(19) Operation speed and read/write margins of a memory cell **125** can be benchmarked by tunnel magnetoresistance ratio (TMR), defined as

$$TMR = ((R_{ap} + R_s) - (R_p + R_s)) / (R_p + R_s) = (R_{ap} - R_p) / (R_p + R_s).$$

Since the routing resistance R_s is in the denominator of the expression of TMR, the routing resistance degrades TMR. Accordingly, additional resistance other than the resistance of the MTJ itself shall be reduced to or kept at a desirable value as low as possible to safeguard sensitivity and speed of the memory cell. There is, however, a large portion of routing resistance due to vias in the memory cell, such as vias connecting the source/drain regions of the transistor **190** to respective source line and bit line. There is a need to reduce routing resistance associated with vias for achieving larger read/write margins and faster read/write operations of the memory cells.

(20) FIG. 3 illustrates a schematic view of a memory array **300**, in accordance with an embodiment. The memory array **300** includes a plurality of memory cells **302**, which may be implemented as the memory cells **125** in FIGS. 1 and 2. Each of the memory cell **302** includes an MTJ **304** illustrated as a free layer FL and a corresponding pinned layer PL for simplicity. As illustrated in FIG. 3, the memory array **300** includes MTJs **304** organized in an array (e.g., in rows and columns), and has bit lines (e.g., BL0, BL1), word lines (e.g., WL0, WL1), and common source lines (e.g., CSL). Each of the MTJ **304** is coupled between a bit line and a drain of a corresponding transistor **306**. A gate of the transistor **306** is coupled to a word line, and a source of the transistor **306** is coupled to a common source line. When a transistor **306** is turned on, a current

flowing through the drain and the source of the transistor **306** is determined by the resistance of the MTJ **304** (e.g., a high resistance R_{ap} or a low resistance R_p), and the current is used to determine whether a “0” or a “1” is stored in the MTJ **304**. As illustrated in FIG. 3, one MTJ **304** is associated with one transistor **306**. In some alternative embodiments, one MTJ **304** may be associated with two or more transistors connected in parallel. The configuration of transistors connected in parallel reduces channel resistance contributed from the transistors. Further, in the example of FIG. 3, four bits are stored by the four MTJs **304**. One skilled in the art will readily appreciate that the memory array **300** may include more MTJs **304** than illustrated in FIG. 3 to store a pre-determined amount of data bits.

(21) FIG. 4 illustrates a schematic cross-sectional view of the memory array **300** of FIG. 3, in accordance with an embodiment. For simplicity, FIG. 4 illustrates only a portion of the memory array **300**, particular the memory cell **302** in the dashed rectangular box shown in FIG. 3.

(22) As illustrated in FIG. 4, the memory cell **302** includes a first transistor **T1** and a second transistor **T2** arranged on a substrate **402**. In one embodiment, the transistors **T1** and **T2** are field-effect transistors (FETs), such as metal-oxide-semiconductor field-effect transistors (MOSFETs). In some embodiments, the transistors **T1** and **T2** are formed as planar FETs or non-planar FETs. In furtherance of some embodiments, each of the transistors **T1** and **T2** is a FinFET device. FinFETs may have one or more non-planar gate structures for wrapping partially or completely around one or more channel regions. As illustrated in FIG. 4, the first transistor **T1** has a gate structure **404G-1** disposed over the substrate **402** between a source region **404S-1** and a drain region **404D**. The second transistor **T2** has a gate structure **404G-2** disposed over the substrate **402** between a source region **404S-2** and the drain region **404D**. The drain region **404D** is a common drain region shared by the transistors **T1** and **T2**. The source regions **404S-1**, **404S-2** and the drain region **404D** are collectively referred to as source/drain regions **404**. Each of the gate structures **404G-1** and **404G-2** includes a gate electrode **408** separated from the substrate **402** by a gate dielectric **406**. In some embodiments, the gate electrode **408** may comprise polysilicon. In such embodiments, the gate dielectric **406** may include a dielectric material, such as an oxide (e.g., silicon dioxide), a nitride (e.g., silicon-nitride), or the like. In other embodiments, the gate electrode **408** may comprise a metal, such as aluminum, copper, titanium, tantalum, tungsten, molybdenum, cobalt, or the like. In such embodiments, the gate dielectric **406** may comprise a high-k dielectric material, such as hafnium oxide, hafnium silicon oxide, hafnium tantalum oxide, aluminum oxide, zirconium oxide, or the like.

(23) Source/drain contacts **MD** are formed over the source/drain regions **404**. Particularly, source contacts **MD-S** are formed over the source regions **404S-1** and **404S-2** in an interlayer dielectric (ILD) layer. Drain contact **MD-D** is formed over the common drain region **404D** in the ILD layer. A plurality of inter-metal dielectric (IMD) layers (e.g., **IMD0-IMD6**) are formed over the ILD layer, with each IMD layer having conductive lines (e.g., **M0-M6**) and vias (e.g., **VD** and **V0-V5**) formed therein. In the example of FIG. 4, vias **VD** connect the source contacts **MD-S** and the drain contact **MD-D** to respective conductive lines **M0** formed in the IMD layer **IMD0**. Similarly, gate vias **VG** connect the gate structures **414G-1** and **414G-2** to respective conductive lines **M0** formed in the IMD layer **IMD0**. The via **V0** connects the conductive line **M0** to the conductive line **M1** formed in the IMD layer **IMD1**. The via **V1** connects the conductive line **M1** to the conductive line **M2** formed in the IMD layer **IMD2**. The via **V2** connects the conductive line **M2** to the conductive line **M3** formed in the IMD layer **IMD3**. The via **V3** connects the conductive line **M3** to the conductive line **M4** formed in the IMD layer **IMD4**. In the example of FIG. 4, the MTJ structure **420** is formed in the IMD layer **IMD5**. The MTJ structure **420** illustrated in FIG. 4 is a simplified schematic view showing a bottom electrode via (BEVA) **422**, an MTJ **424**, and a top electrode via (TEVA) **426**. The BEVA **422** connects the conductive line **M4** to the MTJ **424**. The TEVA **426** connects the MTJ **424** to the via **V5** formed in the IMD layer **IMD6**. The via **V5** connects the TEVA **426** to the conductive line **M6** formed in the IMD layer **IMD6**. The two conductive lines **M0**

coupled to the two source regions **404S-1** and **404S-2** are further coupled together, forming the common source line (CSL). The two conductive lines **M1** coupled to the two gate structures **404G-1** and **404G-2** are further coupled together, as the word line (WL). Accordingly, the transistors **T1** and **T2** are connected in parallel. The parallel configuration of the transistors **T1** and **T2** reduces channel resistance by half when the transistors **T1** and **T2** are turned on. While the channel resistance can be reduced with parallel configuration of the transistors, vias **VD** may still serve as a major contributor to the routing resistance, particularly due to generally small cross-sectional areas of this type of vias.

(24) In the illustrated embodiment in FIG. 4, the word lines WL (e.g., **WL0**, **WL1**) of the memory array are formed in the IMD layer **IMD1**, the common source line CSL are formed in the IMD layer **IMD0**, the bit lines BL (e.g., **BL0**, **BL1**) are formed in the IMD layer **IMD6**, and the MTJ structure is formed in the IMD layer **IMD5**. These are, of course, merely examples and non-limiting. The word lines, bit lines, common source line, and the MTJ structure may be formed in other IMD layers, these and other variations are fully intended to be included within the scope of the present disclosure.

(25) FIG. 5 is a layout or a top view **500** of a portion of a memory array comprising MTJs, in accordance with some embodiments. In some embodiments, the memory array includes gate structures **404G-1**, **404G-2**, **404G-3**, **404G-4** (collectively, gate structures **404G**) elongated along the Y-direction, active regions **430-1**, **430-2** (collectively, active regions **430**) elongated along the X-direction. These components may be arranged and function as the memory array **300** described above with respect to FIG. 3. In one aspect, the memory array **300** includes more, fewer, or different components than shown in FIG. 5. For example, the memory array **300** includes additional components (e.g., routing metals, via contacts) that are not shown in FIG. 5.

(26) In the illustrated embodiment, transistors **T1-T8** are formed where the gate structures **404G-1**, **404G-2**, **404G-3**, **404G-4** and active regions **430-1**, **430-2** intersect. For example, a transistor **T1** is formed, at which the active region **430-1** and the gate structure **404G-1** intersect. For example, a transistor **T2** is formed, at which the active region **430-1** and the gate structure **404G-2** intersect. In some embodiments, the transistors **T1** and **T2** are formed as planar FETs or non-planar FETs. In furtherance of some embodiments, each of the transistors **T1** and **T2** is a FinFET device. The transistor **T1** includes a source region **404S-1** and a drain region **404D**. The transistor **T2** includes a source region **404S-2** and the drain region **404D**. The transistors **T1** and **T2** share the drain region **404D**. The source regions and common drain regions of the other transistors **T3-T8** are similarly disposed in the layout **500** and not reiterated herein for the sake of conciseness.

(27) The memory array includes source contacts **MD-S-1**, **MD-S-2**, **MD-S-3** (collectively, source contacts **MD-S**) elongated along the Y-direction. Each of the source contact **MD-S** extends across the active regions **430-1** and **430-2** and in contact with the source regions formed in the active regions **430-1** and **430-2**. Accordingly, sources regions associated with the same gate structure in different active regions are coupled together through the respective source contact **MD-S**. In one example, each of the source contacts **MD-S** has a width along the X-direction from about 15 nm to about 25 nm. The memory array also includes drain contacts **MD-D-1**, **MD-D-2**, **MD-D-3**, **MD-D-4** (collectively, drain contacts **MD-D**) elongated along the Y-direction. Each of the drain contact **MD-D** extends across the respective common drain region. For example, the drain contact **MD-D-1** is in contact with the common drain region of the transistors **T1** and **T2**, the drain contact **MD-D-2** is in contact with the common drain region of the transistor **T3** and **T4**, the drain contact **MD-D-3** is in contact with the common drain region of the transistors **T5** and **T6**, and the drain contact **MD-D-4** is in contact with the common drain region of the transistors **T7** and **T8**. In one example, each of the drain contact **MD-D** has a length along the Y-direction from about 35 nm to about 50 nm, and a width along the X-direction from about 15 nm to about 25 nm.

(28) The memory array includes conductive lines **M0-1**, **M0-2**, **M0-3**, **M0-4**, **M0-5**, **M0-6**, **M0-7** (collectively, conductive lines **M0**) formed in the IMD layer **IMD0** and elongated along the X

direction. The conductive line **M0-1** extends across the gate structures **404G-1**, **404G-2**, **404G-3**, **404G-4**. The gate via **VG-1** connects the gate structure **404G-1** to the conductive line **M0-1**. The gate via **VG-2** connects the gate structure **404G-2** to the conductive line **M0-1**. Not depicted in FIG. 5, the conductive line **M0-1** is further coupled to a conductive line **M1** as a word line formed in the IMD layer **IMD1**, such as illustrated in FIG. 4.

(29) The conductive line **M0-2** extends across the source contacts **MD-S-1**, **MD-S-2**, **MD-S-3**. The conductive line **M0-2** serves as a common source line (CSL). Other than relying on multiple vias **VD** individually connecting each source contact to the conductive line **M0-2**, a via rail **VDR-1** is formed between the source contacts **MD-S-1**, **MD-S-2**, **MD-S-3** and the conductive line **M0-2**. The via rail **VDR-1** connects each of the source contacts **MD-S-1**, **MD-S-2**, **MD-S-3** to the conductive line **M0-2**. In the illustrated embodiment, the via rail **VDR-1** has a length along the X-direction same as the conductive line **M0-2** and a width along the Y-direction smaller than the conductive line **M0-2**. In one example, the conductive line **M0-2** has a width from about 25 nm to about 35 nm, and the via rail **VDR-2** has a width from about 15 nm to about 20 nm. By having one continuous via rail other than multiple individual vias, the contact area between the via and the conductive line **M0-2** and the contact area between the via and the source contacts **MD-S** are both increased, which leads to smaller routing resistance for the common source line. In some instances, the routing resistance for the common source line may be reduced for about 15% by implementing the via rail. Similarly, the conductive line **M0-3** extends across the source contacts **MD-S-1**, **MD-S-2**, **MD-S-3**, and the via rail **VDR-2** connects each of the source contacts **MD-S-1**, **MD-S-2**, **MD-S-3** to the conductive line **M0-3**.

(30) Each of the conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7** extends across the respective drain contact **MD-D**. The conductive lines **M0-4** and **M0-7** also overlap with the source contact **MD-S-2** in a top view. The conductive line **M0-5** overlaps with the source contact **MD-S-1** in a top view. The conductive line **M0-6** overlaps with the source contact **MD-S-3** in a top view. In one example, each of the conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7** has a length along the X-direction from about 70 nm to about 95 nm, and a width along the W-direction from about 12 nm to about 25 nm. That is, the width of the conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7** is smaller than the width of the conductive line **M0-2**. Multiple vias **VD-1**, **VD-2**, **VD-3**, **VD-4** (collectively, vias **VD**) connect the drain contacts **MD-D** to the respective conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7** thereabove. Particularly, the via **VD-1** connects the drain contact **MD-D-1** to the conductive line **M0-4**, the via **VD-2** connects the drain contact **MD-D-2** to the conductive line **M0-6**, the via **VD-3** connects the drain contact **MD-D-3** to the conductive line **M0-5**, and the via **VD-4** connects the drain contact **MD-D-4** to the conductive line **M0-7**. Each of the vias **VD** has an extended width along the X-direction that is larger than the width of the respective drain contact **MD-D** along the X-direction. The extended width increases the contact area between the vias **VD** and the respective drain contact **MD-D** and helps reducing routing resistance. The vias **VD** with an extended width are also referred to as slot vias **VD**. In the illustrated embodiment, the width of the vias **VD** along the Y-direction is substantially the same as the width of the conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7**. In some embodiments, the slot via **VD** has a square space in a top view with extended width on all four sides. In some alternative embodiments, the slot via **VD** has a rectangular shape in a top view, as shown in FIG. 5. The conductive lines **M0-4**, **M0-5**, **M0-6**, **M0-7** couple the drain contacts **MD-D** to MTJs formed in a higher IMD layer (e.g., **IMD5**). The four MTJs overlying four associated drain contacts **MD-D** in the layout **500** are represented by four dashed square boxes in FIG. 5.

(31) Reference is now made to FIGS. **6A**, **6B**, **6C**, collectively. FIG. **6A** illustrates a cross-sectional view taken in the X-Z plane along the A-A line of the portion of the memory array shown in FIG. 5. FIG. **6B** illustrates a cross-sectional view taken in the X-Z plane along the B-B line of the portion of the memory array shown in FIG. 5. FIG. **6C** illustrates a cross-sectional view taken in the Y-Z plane along the C-C line of the portion of the memory array shown in FIG. 5.

(32) In some embodiments, the substrate **402** may be a semiconductor substrate such as a silicon

substrate. The substrate **402** may include various layers, including conductive or insulating layers formed on a semiconductor substrate. The substrate **402** may include various doping configurations depending on design requirements as is known in the art. For example, different doping profiles (e.g., n-wells, p-wells) may be formed on the substrate **402** in regions designed for different device types (e.g., n-type field effect transistors (NFET), p-type field effect transistors (PFET)). The suitable doping may include ion implantation of dopants and/or diffusion processes. The substrate **402** may have isolation features (e.g., shallow trench isolation (STI) features) interposing the regions providing different device types. The substrate **402** may also include other semiconductors such as germanium, silicon carbide (SiC), silicon germanium (SiGe), or diamond. Alternatively, the substrate **402** may include a compound semiconductor and/or an alloy semiconductor. Further, the substrate **402** may optionally include an epitaxial layer (epi-layer), may be strained for performance enhancement, may include a silicon-on-insulator (SOI) structure, and/or may have other suitable enhancement features.

(33) In some embodiments, the active regions **430-1** and **430-2** are fin-like structure designed to form fin-like field effect transistors (FinFETs). The active regions **430-1** and **430-2** may protrude from the substrate **402** and extend in parallel in the X-direction. The fin-like structure may be formed by patterning the substrate **402** using one or more photolithography processes, including double-patterning or multi-patterning processes. The active regions **430-1** and **430-2** are separated by isolation structure **410**. The isolation structure **410** may include silicon oxide, silicon nitride, silicon oxynitride, other suitable isolation material (for example, including silicon, oxygen, nitrogen, carbon, or other suitable isolation constituent), or combinations thereof. The isolation structure **230** may include different structures, such as shallow trench isolation (STI) structures, deep trench isolation (DTI) structures, and/or local oxidation of silicon (LOCOS) structures.

(34) Gate structures **404G** are formed over the substrate **402** and across the active regions **430**. Each of the gate structures **404G** includes a gate stack having a gate dielectric and a gate electrode disposed on the gate dielectric. The gate dielectric includes a dielectric material, such as silicon oxide, germanium oxide, high k dielectric material layer or a combination thereof. In another embodiment, the gate dielectric includes an interfacial layer (such as a silicon oxide or germanium oxide layer) and a high-k dielectric material layer on the interfacial layer. The gate electrode includes a conductive material layer, such as doped polycrystalline silicon (polysilicon), metal, metal alloy or combinations thereof. The gate structures **404G** may be formed by a procedure that includes forming a gate dielectric layer, forming a gate electrode layer on the gate dielectric layer, and patterning the gate electrode layer and the gate dielectric layer. The formation of the gate structures **404G** may further include a gate replacement procedure to replace the previously formed gate stack with high-k dielectric and metal. The gate replacement may include a gate last operation or a high-k last operation where both gate dielectric and gate electrode are replaced at a later fabrication stage. The gate structures **404G** may also include gate spacers formed on sidewalls of the gate structures **404G** by a procedure that includes deposition and anisotropic etch.

(35) In some embodiments, each of the source regions **404S** and drain regions **404D** includes an epitaxial source/drain feature formed over the active region **430-1** or **430-2** in the respective source region or drain region. For NFETs, the epitaxial source/drain features may be of n-type doped. For PFETs, the epitaxial source/drain features may be of p-type doped. For example, for NFETs, the epitaxial source/drain features may include silicon and be doped with carbon, phosphorous, arsenic, other n-type dopant, or combinations thereof; and for PFETs, the epitaxial source/drain features may include silicon germanium or germanium and be doped with boron, other p-type dopant, or combinations thereof. The epitaxial source/drain features may be formed by epitaxially growing semiconductor material(s) (e.g., Si, SiGe) over the active regions **430-1** and **430-2**, for example, using CVD deposition techniques (e.g., Vapor Phase Epitaxy), molecular beam epitaxy, other suitable epitaxial growth processes, or combinations thereof.

(36) The ILD layer is formed over the substrate **402** and the gate structures **404G**. The IMD0 layer

is formed over the ILD layer. The ILD layer and the IMD0 layer may be formed of any suitable dielectric material, for example, a nitride such as silicon nitride; an oxide such as silicon oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), or the like; or the like. The ILD layer and the IMD0 layers may be formed by any acceptable deposition process, such as spin coating, physical vapor deposition (PVD), chemical vapor deposition (CVD), the like, or a combination thereof.

(37) Each of the source contacts MD-S continuously extends in the Y-direction across multiple source regions **404S** formed in active regions **430-1** and **430-2**. Each of the drain contacts MD-D extends in the Y-direction across the respective drain region **404D** but does not extend to adjacent other active regions. In some embodiments, the source contacts MD-S and the drain contacts MD-D are formed by forming trenches in the ILD layer and fill the trenches with conductive materials, such as titanium nitride (TiN), tantalum (Ta), titanium (Ti), tantalum nitride (TaN), ruthenium (Ru), tungsten (W), cobalt (Co), aluminum (Al), molybdenum (Mo), titanium silicide (TiSi), tungsten silicon (WSi), platinum silicide (PtSi), cobalt silicide (CoSi), nickel silicide (NiSi), or a combination thereof. A chemical mechanical polishing (CMP) process may be performed subsequently to remove excessive conductive materials and expose the ILD layer.

(38) The via rails VDR-1 and VDR-2 extends continuously in the X-direction and couples multiple source contacts MD-S together. In some embodiments, the via rails VDR-1 and VDR-2 are formed by forming trenches in the IMD0 layer and fill the trenches with conductive materials, such as titanium nitride (TiN), tantalum (Ta), titanium (Ti), tantalum nitride (TaN), ruthenium (Ru), tungsten (W), cobalt (Co), aluminum (Al), molybdenum (Mo), titanium silicide (TiSi), tungsten silicon (WSi), platinum silicide (PtSi), cobalt silicide (CoSi), nickel silicide (NiSi), or a combination thereof. The vias VD-1 and VD-2 individually land on respective drain contacts MD-D. In some embodiments, the vias VD-1 and VD-2 are formed by forming slots in the IMD0 layer and fill the slots with conductive materials. The conductive material for vias VD-1 and VD-2 may be similar to the via rails VDR-1 and VDR-2. In some embodiments, the via rails VDR-1, VDR-2 and the vias VD-1, VD-2 are formed jointly with the conductive lines M0 in the IMD0 layer using a damascene or dual damascene process. The conductive lines M0 may employ conductive materials such as cobalt (Co), aluminum (Al), copper (Cu), tungsten (W), or a combination thereof.

(39) Referring to FIG. 6A, by having a continuous via rail across multiple source contacts other than isolated vias over each source contact, the contact area is increased between the vias and source contacts, as well as between the vias and the conductive line M0. Accordingly, the routing resistance along the path of the common source line is reduced. Referring to FIG. 6B, the via VD has an extended width $W_{sub.VD}$ measured at its bottom portion, which is larger than a width $W_{sub.MD}$ of the drain contact MD-D measured at its top portion. Accordingly, the whole top surface of the drain contact MD-D is utilized for making contact with the via VD and contributes for a reduced contact resistance, which also helps reducing a portion of the routing resistance in the memory cell. Further, the extended width $W_{sub.VD}$ is larger than the width $W_{sub.MD}$ for about 5% to about 30%. If it is less than about 5%, overlay inaccuracy may cause misalignment between the via VD and the drain contact MD-D; if it is larger than about 30%, the via VD may become too wide and overshadow the adjacent gate structures **404G**, which may impact functions of the gate structures or cause electrical shorting. In some embodiments, edges of the via VD is offset from the gate structures **404G**. In some alternative embodiments, edges of the via VD is directly above the adjacent gate structures **404G**, as illustrated in FIG. 6B. As discussed above, the extended width $W_{sub.VD}$ being no larger than 30% of the width $W_{sub.MD}$ provides a compromise between size of the via VD and performance of the gate structures **404G**. Referring to FIG. 6C, the via rail VDR has a width $W_{sub.VDR}$ measured at its top portion, which is smaller than a width $W_{sub.M0}$ of the conductive line M0 measured at its bottom portion. In some embodiments, the width $W_{sub.VDR}$ is about 40% to about 60% smaller than the width $W_{sub.M0}$. If the width $W_{sub.VDR}$ is more than 60% smaller than the width $W_{sub.M0}$, the via rail VDR may become too narrow and lead to higher

contact resistance. If the width $W_{sub.VDR}$ is less than 40% smaller than the width $W_{sub.M0}$, the via rail VDR may become too wide and may accidentally short to adjacent conductive lines **M0** for drain connections.

(40) FIGS. 7A and 7B illustrate an alternative embodiment to the cross-sectional views as shown in FIGS. 6B and 6C. FIG. 7A illustrates a cross-sectional view taken in the X-Z plane along the B-B line of the portion of the memory array shown in FIG. 5. FIG. 7B illustrates a cross-sectional view taken in the Y-Z plane along the C-C line of the portion of the memory array shown in FIG. 5. Reference numerals are repeated for ease of understanding and similar aspects are not repeated below in the interest of conciseness. One difference between the embodiments in FIG. 7A and FIG. 6B is that the via VD has a reversed-trapezoid shape, allowing a larger top surface in contact with the conductive line **M0**, while maintaining the bottom surface with an extended width $W_{sub.VD}$ not larger than 30% of the width $W_{sub.MD}$. The larger top surface of the via VD increases contact area and reduces contact resistance. The reversed-trapezoid shape may be formed by a first etching process that forms tapered sidewalls during slot formation. One difference between the embodiments in FIG. 7B and FIG. 6C is that the via rail VDR has a trapezoid shape, allowing a larger bottom surface in contact with the source contact MD-S, while maintaining the top surface with a width $W_{sub.VDR}$ at least 40% smaller than the width $W_{sub.M0}$. The larger bottom surface of the via rail VDR increases contact area and reduces contact resistance. The trapezoid shape may be formed by a second etching process that expands lower portion of the trench during trench formation. The first etching process and the second etching process may be performed separately to allow the via VD with a reversed-trapezoid shape and the via rail VDR with a trapezoid shape coexist in one structure.

(41) FIG. 8 illustrates a flow chart of a method **800** for forming a semiconductor device having an MRAM array. Many aspects of the semiconductor device are the same as or similar to those of the memory system **100** illustrated in FIG. 1. While method **800** is illustrated and described below as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events are not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. In addition, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein. Further, one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(42) At operation **802**, active regions are formed on a substrate, such as active regions **430** in FIG. 5. At operation **804**, gate structures are formed across the active regions, such as the gate structures **404G** in FIG. 5. At operation **806**, source/drain regions are formed in active regions located on both sides of the gate structures, such as the source regions **404S** and drain regions **404D** in FIG. 5. At operation **808**, an ILD layer is formed over the gate structures and the source/drain regions, such as the ILD layer in FIGS. 6A-6C. At operation **810**, source contacts and drain contacts are formed in the ILD layer, such as the source contacts MD-S and the drain contacts MD-D in FIG. 5. At operation **812**, an IMD0 layer is formed over the ILD layer, such as the IMD0 layer in FIGS. 6B-6C. At operation **814**, slot vias are formed over the drain contacts and via rails are formed over the source contacts, such as the slot vias VD and via rails VDR in FIG. 5. At operation **816**, conductive lines are formed in the IMD0 layer, such as the conductive lines **M0** in FIG. 5. The conductive lines **M0** overlapping the via rails VDR provide a common source line for the memory array. At operation **818**, other IMD layers are formed over the IMD0 layer and an MTJ structure is formed in one of the higher IMD layers, such as the IMD1~IMD6 and the MTJ structure **420** in FIG. 4.

(43) Although not intended to be limiting, one or more embodiments of the present disclosure provide many benefits to a semiconductor device and the formation thereof. For example, embodiments of the present disclosure provide a semiconductor device with an array of MRAM cells having MTJs. Via rails and slot vias have been implemented to reduce routing resistance and increase sensitivity and speed of the MRAM cells. Furthermore, formation of this semiconductor

device can be readily integrated into existing semiconductor fabrication processes.

(44) In one exemplary aspect, the present disclosure is directed to a memory device. The memory device includes a transistor having a first source/drain (S/D) region and a second S/D region, a first S/D contact disposed over the first S/D region, the first S/D contact extending lengthwise in a first direction, a second S/D contact disposed over the second S/D region, a first via landing on the first S/D contact, the first via extending lengthwise in a second direction different from the first direction, a second via landing on the second S/D contact, the first via having a length measured in the second direction that is larger than the second via, a first conductive line coupled to the first via, a second conductive line coupled to the second via, and a memory structure disposed above the transistor and coupled to the second conductive line. In some embodiments, the memory structure is a magnetic tunnel junction (MTJ). In some embodiments, the MTJ has a bottom electrode coupled to the second conductive line. In some embodiments, the first conductive line is directly above the first via, and the first conductive line extends lengthwise in the second direction. In some embodiments, the first conductive line has a width measured in the first direction larger than that of the first via. In some embodiments, the first S/D region is a source region, the second S/D region is a drain region, the first conductive line couples to a common source line of the memory device, and the second conductive line couples to a bit line of the memory device. In some embodiments, the transistor has a length measured in the second direction from an outer edge of the first S/D region to an outer edge of the second S/D region, and wherein the length of the first via is larger than the length of the transistor. In some embodiments, the second via has a width measured in the second direction that is larger than a width of the second S/D contact. In some embodiments, the width of the second via is larger than the width of the second S/D contact for about 5% to about 30%. In some embodiments, the first via and the first conductive line have a same length measured in the second direction.

(45) In another exemplary aspect, the present disclosure is directed to a memory device. The memory device includes an active region having a first source region, a second source region, and a drain region sandwiched between the first and second source regions, a first contact coupled to the first source region and a second contact coupled to the second source region, each of the first and second contacts extending lengthwise along a first direction, a via extending lengthwise along a second direction that is different from the first direction, the via being in contact with the first and second contacts, a conductive line extending lengthwise along the second direction and coupled to the via, and a magnetic tunnel junction (MTJ) disposed above the active region, wherein the MTJ has an electrode coupled to the drain region. In some embodiments, the conductive line is directly above the via. In some embodiments, the via has a width measured in the first direction that is smaller than the conductive line. In some embodiments, the width of the via is about 40% to about 60% smaller than the conductive line. In some embodiments, the via is a first via, and the memory device further includes a third contact coupled to the drain region, and a second via in contact with the third contact, the second via having a width measured in the second direction that is larger than the third contact. In some embodiments, the first via is wider than the second via in the second direction.

(46) In another exemplary aspect, the present disclosure is directed to a method. The method includes forming an active region on a substrate, forming a first source region and a second source region in the active region, forming a first contact over the first source region and a second contact over the second source region, forming a dielectric layer over the first and second contacts, forming a trench in the dielectric layer, the trench extending continuously from the first contact to the second contact in a top view and exposes the first contact and the second contact, filling the trench with a conductive material, thereby forming a via rail, forming a conductive line in the dielectric layer, the conductive line being in contact with the via rail, and forming a memory structure above the conductive line. In some embodiments, the first and second contacts extend lengthwise in a first direction, and the via rail and the conductive line extend lengthwise in a second direction

perpendicular to the first direction. In some embodiments, the conductive line fully covers the via rail in the top view. In some embodiments, the method further includes forming a drain region in the active region, the drain region being between the first source region and the second source region, forming a third contact over the drain region, and forming a via in contact with the third contact, the via having a width larger than the third contact.

(47) The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand the aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A memory device, comprising: a transistor having an active region that includes a first source/drain (S/D) region and a second S/D region; a first S/D contact disposed over the first S/D region, wherein the first S/D contact extends lengthwise in a first direction and the active region extends lengthwise in a second direction different from the first direction, wherein along the first direction the first S/D contact extends beyond a boundary of the active region in a top view of the memory device; a second S/D contact disposed over the second S/D region; a first via disposed on the first S/D contact, wherein the first via extends lengthwise in the second direction different from the first direction; a second via disposed on the second S/D contact, wherein the first via has a length measured in the second direction that is larger than the second via; a first conductive line coupled to the first via; a second conductive line coupled to the second via; and a memory structure disposed above the transistor and coupled to the second conductive line.
2. The memory device of claim 1, wherein the memory structure is a magnetic tunnel junction (MTJ).
3. The memory device of claim 2, wherein the MTJ has a bottom electrode coupled to the second conductive line.
4. The memory device of claim 1, wherein the first conductive line is directly above the first via, and the first conductive line extends lengthwise in the second direction.
5. The memory device of claim 4, wherein the first conductive line has a width measured in the first direction larger than that of the first via.
6. The memory device of claim 1, wherein the first S/D region is a source region, the second S/D region is a drain region, the first conductive line couples to a common source line of the memory device, and the second conductive line couples to a bit line of the memory device.
7. The memory device of claim 1, wherein the transistor has a length measured in the second direction from an outer edge of the first S/D region to an outer edge of the second S/D region, and wherein the length of the first via is larger than the length of the transistor.
8. The memory device of claim 1, wherein the second via has a width measured in the second direction that is larger than a width of the second S/D contact.
9. The memory device of claim 8, wherein the width of the second via is larger than the width of the second S/D contact for about 5% to about 30%.
10. The memory device of claim 1, wherein the first via and the first conductive line have a same length measured in the second direction.
11. A memory device, comprising: an active region having a first source region, a second source region, and a drain region sandwiched between the first and second source regions; a first gate structure disposed between the first source region and the drain region; a second gate structure

disposed between the second source region and the drain region; a first contact coupled to the first source region and a second contact coupled to the second source region, wherein each of the first and second contacts extends lengthwise along a first direction; a via extending lengthwise along a second direction that is different from the first direction, wherein the via is in contact with the first and second contacts; a conductive line extending lengthwise along the second direction and across both the first and second gate structures in a top view of the memory device, wherein the conductive line interfaces a top surface of the via; and a magnetic tunnel junction (MTJ) disposed above the active region, wherein the MTJ has an electrode coupled to the drain region.

12. The memory device of claim 11, wherein the conductive line is directly above the via.

13. The memory device of claim 11, wherein the via has a width measured in the first direction that is smaller than the conductive line.

14. The memory device of claim 13, wherein the width of the via is about 40% to about 60% smaller than the conductive line.

15. The memory device of claim 11, wherein the via is a first via, further comprising: a third contact coupled to the drain region; and a second via in contact with the third contact, wherein the second via has a width measured in the second direction that is larger than the third contact.

16. The memory device of claim 15, wherein the first via is wider than the second via in the second direction.

17. A semiconductor device, comprising: a continuous active region having a first source/drain (S/D) region, a second S/D region, and a third S/D region sandwiched between the first and second S/D regions along a first direction; a first contact landing on the first S/D region; a second contact landing on the second S/D region, wherein each of the first and second contacts extends lengthwise along a second direction perpendicular to the first direction; a first via extending lengthwise along the first direction and coupled to the first and second contacts; a third contact landing on the third S/D region; a second via coupled to the third contact, wherein the first via has a length measured in the first direction that is larger than the second via; and a magnetic tunnel junction (MTJ) coupled to the second via.

18. The semiconductor device of claim 17, wherein the MTJ is disposed directly above the third S/D region.

19. The semiconductor device of claim 17, further comprising: a first gate structure disposed between the first and third S/D regions; and a second gate structure disposed between the second and third S/D regions, wherein the first and second gate structures are directly under the first via.

20. The semiconductor device of claim 17, wherein in a cross-sectional view along the second direction, the first via has a top surface narrower than a bottom surface.
